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Simple Protection Techniques

Snubber Circuit Basics for SCR and MOSFET

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Outline

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SCR Snubber Protection

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What is a Snubber Circuit? ⚡

Definition

A snubber is a small auxiliary network, typically resistor–capacitor (RC) or resistor–capacitor–diode (RCD), connected across a power semiconductor device to absorb transient electrical energy and shape the voltage or current the device experiences during switching.

Purpose of a snubber:

- Limit the rate of voltage rise (dv/dt) across the device
- Limit the rate of current rise (di/dt) through the device
- Clamp voltage overshoot and spikes at turn-off
- Reduce electromagnetic interference (EMI)
- Keep the device inside its Safe Operating Area (SOA)

Why Protection is Needed

- Real circuits always contain parasitic elements: stray wiring inductance, transformer leakage inductance, and device junction capacitance.
- During switching transitions, energy stored in these parasitic elements produces fast voltage and current transients.
- Two classic failure modes:
 - ⊗ **False / unwanted turn-on** due to high dv/dt – mainly affects latching devices such as the SCR.
 - ⊗ **Destructive overvoltage** at turn-off due to di/dt through stray or leakage inductance – mainly affects fast switches such as the MOSFET.
- A correctly designed snubber keeps the device within its datasheet ratings and extends its operating life.

Classification of Snubber Circuits

➤ **Unpolarized RC snubber**

Simple resistor–capacitor pair, used mainly for dv/dt limiting (typical with SCR and TRIAC).

➤ **Polarized RCD snubber**

Adds a diode to direct energy into the capacitor, with a resistor for controlled discharge; used mainly for turn-off voltage clamping (typical with MOSFET and IGBT).

➤ **Turn-on snubber (series inductor, RL type)**

Limits di/dt at turn-on. Mentioned here for completeness, not covered in depth in this session.

➤ **Turn-off snubber (RCD type)**

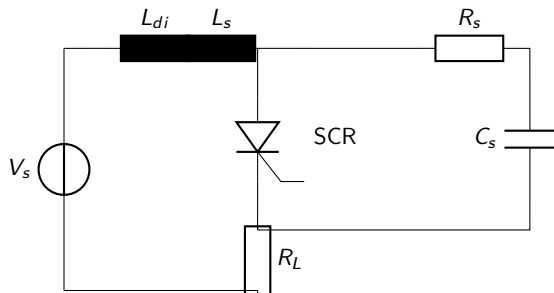
Limits dv/dt and clamps voltage overshoot at turn-off.

This session focuses on the RC snubber for SCR and the RCD snubber for MOSFET.

SCR Switching Behaviour – A Quick Recap

- **Turn-on:** A gate pulse triggers the SCR once it is forward biased. Once latched on, the gate loses control – the device stays on until current falls below the holding value.
- **Unwanted turn-on:** A sufficiently high forward dv/dt , even without a gate pulse, can turn the SCR on through displacement current in the internal junction capacitance.
- **Turn-off:** By natural or forced commutation. Immediately after commutation, a reapplied voltage appears across the device – this is the critical stress window that the snubber must manage.

SCR Snubber – Circuit Diagram



Topology: an unpolarized RC network (R_s in series with C_s) is connected directly across the anode–cathode terminals of the SCR (nodes A–B), in parallel with the device, to limit dv/dt .

- L_{di} – a small series inductor limiting di/dt at turn-on (a separate, deliberately added protection element).
- L_s – the stray/loop inductance of the source and wiring, not an added

SCR Snubber – Design Equations

1) Capacitor selection (dv/dt criterion):

$$C_s \geq \frac{I_L}{\left(\frac{dv}{dt}\right)_{\text{crit}}}$$

I_L : load current; $(dv/dt)_{\text{crit}}$: critical reapplied voltage rate from the SCR datasheet.

2) Resistor selection (must satisfy both):

$$R_s \geq \frac{V_s}{I_{TSM}} \text{ (discharge limit),} \quad R_s \geq 2\sqrt{\frac{L_s}{C_s}} \text{ (critical damping)}$$

I_{TSM} : SCR surge current rating. L_s : stray/loop inductance. Take R_s as the *larger* of the two results.

3) Snubber power dissipation (repetitive operation):

$$P_{R_s} = \frac{1}{2} C_s V_s^2 f$$

SCR Snubber – Worked Example

Given:

- › $V_s = 400 \text{ V}$
- › $I_L = 25 \text{ A}$
- › $(dv/dt)_{crit} = 50 \text{ V } \mu\text{s}^{-1}$
- › $I_{TSM} = 150 \text{ A}$
- › $L_s = 8 \mu\text{H}$ (estimated stray/loop inductance)
- › Line frequency $f = 50 \text{ Hz}$

Design:

$$C_s \geq \frac{25}{50} = 0.5 \mu\text{F}$$

$$R_s \geq \frac{400}{150} = 2.67 \Omega \text{ (discharge)}$$

$$R_s \geq 2\sqrt{\frac{8}{0.5}} = 8.0 \Omega \text{ (damping)}$$

Selected: $C_s = 0.5 \mu\text{F}$, $R_s = 8.2 \Omega$

Damping governs here, not the discharge limit.

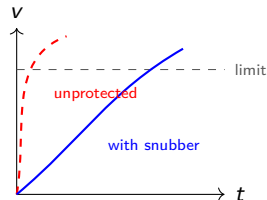
Power Check

$P_{R_s} = 0.5 \times 0.5 \mu\text{F} \times (400)^2 \times 50 = 2 \text{ W} \Rightarrow$ use a 5 W resistor for margin.

(Independent of R_s , provided $R_s C_s \ll$ half the line period – here $4.1 \mu\text{s}$ vs. 10 ms , easily

SCR Snubber – How It Works 💡

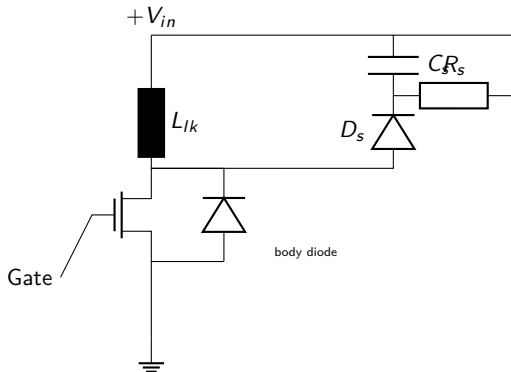
- After the SCR turns off, a reapplied voltage appears across the device.
- Without a snubber, this voltage can rise very fast, charging the internal junction capacitance enough to retrigger the SCR – an unwanted turn-on.
- C_s diverts the fast-changing current, slowing the voltage rise across the SCR.
- R_s limits the discharge current from C_s into the SCR at the next turn-on, and also damps the $L_s C_s$ ringing against stray loop inductance.
- Net effect: the SCR sees a controlled, gentle rise instead of a steep transient



MOSFET Switching Behaviour – A Quick Recap

- MOSFETs turn on and off through the gate voltage and switch very fast, typically nanoseconds to a few microseconds.
- No latching behaviour, unlike the SCR – turn-off is fully controllable through the gate.
- This very speed is what stresses parasitic inductance: at turn-off the current changes rapidly through stray or transformer leakage inductance L_{lk} .
- By $v = L \frac{di}{dt}$, this produces a sharp voltage spike across the drain–source terminals, which can exceed the device's V_{DS} rating and destroy it.

MOSFET RCD Snubber – Circuit Diagram



Topology: an RCD clamp – diode D_s directs the leakage energy into C_s , while R_s (in parallel with C_s) discharges it before the next switching cycle.

► C_s is referenced to $+V_{in}$, not to ground – it only needs to develop $(V_{clamp} - V_{in})$ across itself for the

MOSFET RCD Snubber – Design Equations

1) Energy stored in leakage / stray inductance:

$$E_{leak} = \frac{1}{2} L_{lk} I_{pk}^2$$

2) Clamp capacitor selection:

$$C_s = \frac{E_{leak}}{(V_{clamp} - V_{in}) \Delta V}$$

V_{clamp} : target maximum drain voltage; $(V_{clamp} - V_{in})$ is the voltage the capacitor itself must develop, since C_s is referenced to $+V_{in}$, not ground. ΔV : allowed ripple on C_s , typically 5 %–10 % of $(V_{clamp} - V_{in})$.

3) Resistor selection (full discharge within one cycle):

$$R_s \leq \frac{T_{sw}}{5 C_s}, \quad T_{sw} = \frac{1}{f_{sw}}$$

4) Snubber power dissipation:

$$P_{R_s} = E_{leak} \times f_{sw}$$

MOSFET RCD Snubber – Worked Example

Given:

- › $V_{in} = 48 \text{ V}$
- › $I_{pk} = 10 \text{ A}$
- › $L_{lk} = 5 \mu\text{H}$
- › $V_{clamp} = 100 \text{ V}$
- › $\Delta V = 10 \text{ V}$
- › $f_{sw} = 100 \text{ kHz}$

Design:

$$E_{leak} = \frac{1}{2}(5)(10)^2 = 250 \mu\text{J}$$

$$C_s = \frac{250}{(100 - 48) \times 10} \approx 0.48 \mu\text{F}$$

$$R_s \leq \frac{10 \mu\text{s}}{5 \times 0.47} \approx 4.26 \Omega$$

Selected: $C_s = 0.47 \mu\text{F}$, $R_s = 3.9 \Omega$

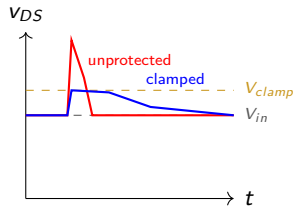
$(V_{clamp} - V_{in}) = 52 \text{ V}$ is the voltage C_s must develop.

Note on Losses

$P_{R_s} = 250 \mu\text{J} \times 100 \text{ kHz} = 25 \text{ W}$ – RCD snubbers are simple but lossy; best suited to low-to-moderate power or protection-priority designs.

MOSFET Snubber – How It Works

- At turn-off, current in L_{lk} cannot change instantly and forces a voltage spike at the drain node.
- Diode D_s conducts once the drain voltage exceeds the clamp level, steering the spike energy into C_s instead of the MOSFET.
- C_s absorbs this energy, holding the drain voltage near the clamp level V_{clamp} .
- R_s bleeds the stored energy out of C_s before the next switching cycle begins.
- Net effect: the MOSFET sees a clamped, bounded voltage instead of an unbounded spike.



- Keep snubber leads and PCB traces as short as possible – extra length adds stray inductance and defeats the purpose.
- Use a fast-recovery diode for the RCD snubber's diode branch; a slow diode will not clamp the spike in time.
- Add margin on resistor power rating – roughly twice the calculated dissipation is a safe rule of thumb.
- Check the RC time constant against the switching period so the snubber fully resets before the next cycle.
- Treat the calculated values as a starting point – always validate with bench measurements (double-pulse test, oscilloscope) before finalizing.

Comparison: SCR Snubber vs. MOSFET Snubber

Parameter	SCR (RC Snubber)	MOSFET (RCD Snubber)
Primary concern	False turn-on via dv/dt	Turn-off voltage spike via di/dt
Topology	R_s in series with C_s , across device	D_s in series with C_s ; R_s in parallel with C_s
Diode required	No	Yes (fast recovery)
Typical C_s range	0.1 – 1 μF	Tens of nF to a few μF , power dependent
Damping / discharge role of R_s	Limits SCR turn-on discharge surge	Discharges C_s before next switching cycle
Key design driver	Datasheet $(dv/dt)_{crit}$	Leakage inductance energy

Summary / Key Takeaways ✓

- Snubbers are simple, low-cost circuits that keep power semiconductor devices within their safe operating limits.
- **SCR:** the RC snubber primarily limits reapplied dv/dt , preventing unwanted retriggering.
- **MOSFET:** the RCD snubber primarily clamps the turn-off voltage spike caused by stray or leakage inductance.
- Correct sizing, based on datasheet ratings and known parasitic elements, is essential – an oversized snubber wastes power, an undersized one fails to protect.
- Always validate the design on the bench; the equations here give a sound starting point, not a final answer.

References: M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*; N. Mohan, T. M. Undeland, W. P. Robbins, *Power Electronics: Converters, Applications and Design*.